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 Code



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README



EXP-2-Image-Capture-and-Video-Processing-Using-OpenCV

Aim:

To write a Python program using OpenCV to capture an image from the webcam and perform the following operations:

Write the frame as a JPG file Display the video Display the video by resizing the window Rotate and display the video



Software Used

Anaconda – Python 3.7 Jupyter Notebook / VS Code OpenCV (cv2)



Algorithm:

Step 1: Import the required libraries and initialize the webcam using cv2.VideoCapture().

Step 2: Capture frames continuously from the webcam.

Step 3:import cv2 import matplotlib.pyplot as plt from IPython.display import clear_output import time

```
cap = cv2.VideoCapture(0) ret, frame = cap.read() if ret: cv2.imwrite("captured_frame.jpg", frame)
cap.release()

captured_image = cv2.imread('captured_frame.jpg')

plt.imshow(captured_image[:,::-1]) plt.title('Captured Frame') plt.axis('off') plt.show()

cap = cv2.VideoCapture(0)

for i in range(50): ret, frame = cap.read() if not ret: break frame_rgb = cv2.cvtColor(frame,
cv2.COLOR_BGR2RGB) clear_output(wait=True) plt.imshow(frame_rgb) plt.axis('off') plt.show() time.sleep(0.05)

cap.release()

cap = cv2.VideoCapture(0)

for i in range(50): ret, frame = cap.read() if not ret: break resized_frame = cv2.resize(frame, (100, 150)) #
Resize to 320x240 frame_rgb = cv2.cvtColor(resized_frame, cv2.COLOR_BGR2RGB) clear_output(wait=True)
plt.imshow(frame_rgb) plt.axis('off') plt.show() time.sleep(0.05)

cap.release()

cap = cv2.VideoCapture(0)

for i in range(50): ret, frame = cap.read() if not ret: break rotated_frame = cv2.rotate(frame,
cv2.ROTATE_90_CLOCKWISE) frame_rgb = cv2.cvtColor(rotated_frame, cv2.COLOR_BGR2RGB)
clear_output(wait=True) plt.imshow(frame_rgb) plt.axis('off') plt.show() time.sleep(0.05)

cap.release()
```



Output

i) Write the frame as JPG image:

Captured image :

Captured Frame



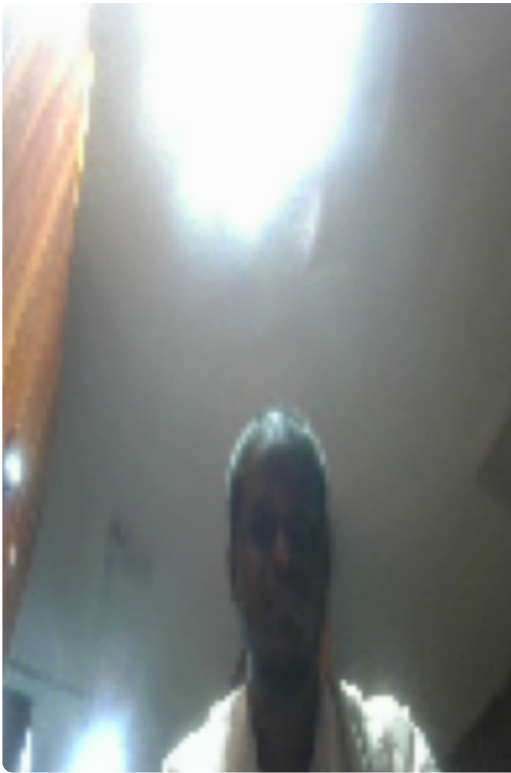
i) Display the video:

Live webcam video:



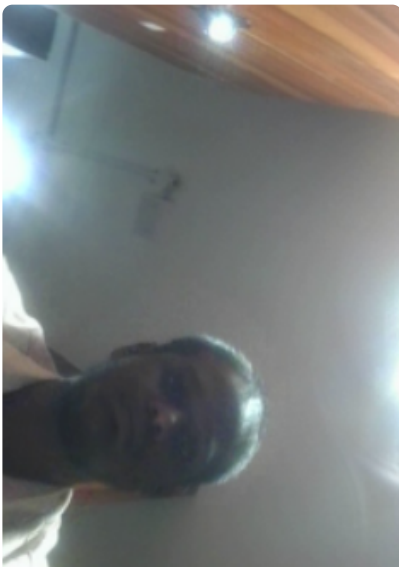
iii) Display the video by resizing the window

Video is shown in resized resolution:



iv) Rotate and display the video

Video is displayed after rotation (90° clockwise):



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